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Capstone Project Report

on

PLC-Based Overall Equipment Effectiveness (OEE) Monitoring System

*Submitted in partial fulfillment of the requirements of the **Third Year** of*

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In

Electronics and Computer Science

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This is to certify that the following students of T. Tech in Electronics and Computer Science Department have satisfactorily completed the requirements for the courses Programmable Logic Controllers and SCADA prescribed by **ST. JOHN COLLEGE OF ENGINEERING AND MANAGEMENT (An Autonomous College affiliated to University of Mumbai)**, while working on "PLC-Based Overall Equipment Effectiveness (OEE) Monitoring System".

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Abstract

Overall Equipment Effectiveness (OEE) is a widely used key performance indicator that shows how effectively a manufacturing machine or production line is being utilised. It combines the losses caused by downtime, reduced speed, and defective output into a single percentage that operators, supervisors, and managers can understand quickly. This project presents a PLC-based OEE Monitoring System that records production activity and stoppage events using digital inputs and computes an OEE-related result inside the PLC.

The system uses five digital inputs: X0 (Start / Production ON), X1 (Stop / Idle), X2 (Minor Stop), X3 (Major Stop), and X4 (Reset). While production is enabled by X0, the PLC increments the Total Production Count (D0), the Idle Count (D1), the Minor Stop Count (D2), and the Major Stop Count (D3). A constant value of 5 is stored in D20 on the first scan. The program then calculates D10 = D3 × D20, D12 = D10 × 100, and D14 = D12 ÷ D0, protected by a divide-by-zero check, and the result is made available in PLC registers for display on an HMI, SCADA screen, or PC.

The ladder diagram is developed and tested using Mitsubishi MELSOFT GX Works3 and GX Simulator. The project is supported by a Data Flow Diagram (Level 0 and Level 1), a system hardware and PLC interfacing block diagram, a flow chart, and the ladder program itself. The work demonstrates the practical use of PLC counters, special relays, arithmetic instructions, and comparison contacts for production monitoring, and it provides a base that can be extended to a full Availability × Performance × Quality OEE calculation.

Keywords: PLC, OEE, Overall Equipment Effectiveness, Ladder Diagram, Production Monitoring, Downtime Counting, GX Works3, HMI/SCADA.

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CHAPTER 1: OVERVIEW

1.1 Introduction

A Programmable Logic Controller (PLC) is an industrial digital controller used to automate machines and electrical systems. Besides controlling outputs, a PLC can also collect data from the machine it controls, which makes it a natural platform for production monitoring.

Overall Equipment Effectiveness (OEE) is a standard measure of manufacturing productivity. In its complete form, OEE is the product of three factors: **Availability** (how much of the planned time the machine actually ran), **Performance** (how fast it ran compared with its ideal speed), and **Quality** (how many of the produced parts were good). OEE = Availability × Performance × Quality.

In this project, a PLC is used to build an OEE Monitoring System. Pushbuttons and field signals represent machine states: Start / Production ON (X0), Stop / Idle (X1), Minor Stop (X2), Major Stop (X3), and Reset (X4). The PLC counts production activity and stoppage events in data registers D0 to D3.

The ladder program then applies a constant (D20 = 5) and arithmetic instructions to calculate the intermediate values D10 and D12 and the final result D14. A divide-by-zero check ensures that the result is calculated only when the Total Production Count is greater than zero.

The system is designed and simulated using Mitsubishi MELSOFT GX Works3 and GX Simulator, and the registers are intended to be displayed on an HMI, SCADA system, or PC.

1.2 Background

In many small and medium factories, machine performance is still tracked manually using paper logs or end-of-shift estimates. These methods are slow, depend on the operator's memory, and often hide small stoppages that together cause large losses.

OEE was developed as part of Total Productive Maintenance (TPM) to expose the major equipment losses in a structured way: breakdowns, setup and adjustment, minor stoppages, reduced speed, start-up rejects, and production rejects. An automated system removes the manual effort and records events as they happen.

A PLC already receives signals from sensors and pushbuttons, so it can count cycles and stoppages with very little extra hardware. The PLC scan cycle makes sure that every input is checked continuously, and built-in special relays such as the one-second clock (SM412) and the always-ON relay (SM400) simplify time-based counting and continuous calculation.

This approach reduces manual reporting, improves the accuracy of production data, and gives a flexible base that can be modified by changing the ladder logic rather than the wiring.

1.3 Importance of the Project

- **Performance visibility:** Converts machine activity and stoppages into numbers that can be seen immediately.
- **Loss identification:** Separates idle time, minor stops, and major stops so that the biggest loss source can be identified.
- **Automatic counting:** Removes manual logging errors by counting events directly in PLC registers.
- **Decision support:** Gives managers a simple calculated value (D14) for comparing shifts, machines, or days.
- **Industrial relevance:** Demonstrates a basic PLC monitoring method used in automated manufacturing.
- **Easy troubleshooting:** Input states and register values can be monitored online in GX Works3 and GX Simulator.

1.4 Perspective of Stakeholders and Customers

- **Machine operators:** Can report stoppages with simple pushbuttons and see the counts update.
- **Maintenance technicians:** Can use the minor and major stop counts to find machines that need attention.
- **Production supervisors:** Can compare the calculated result between shifts and take corrective action.
- **Automation engineers:** Can modify the ladder logic, constants, and formulas to suit different machines.
- **Plant managers and customers:** Gain confidence in delivery capability through reliable, data-based performance reporting.
- **Students:** Can understand counting, special relays, arithmetic instructions, and comparison contacts through practical simulation.

1.5 Objectives and Scope of the Project

Objectives

1. To construct a PLC ladder diagram for an OEE monitoring system.
2. To count total production activity (D0) while the Start / Production ON input (X0) is active.
3. To count idle events (D1), minor stops (D2), and major stops (D3) using separate inputs.
4. To store a constant value (D20 = 5) on the first scan and use it in the calculation.

5. To calculate D10 = D3 × D20, D12 = D10 × 100, and D14 = D12 ÷ D0 with divide-by-zero protection.
6. To provide a Reset input (X4) that clears all counters and to simulate and verify the program using Mitsubishi GX Works3 and GX Simulator.

Scope

The project covers the development and simulation of a PLC-based OEE monitoring system. It includes input assignment, register allocation, ladder logic programming, arithmetic calculation, an HMI/SCADA display plan, and functional testing.

The current scope is limited to the PLC logic. The calculation implemented in the supplied ladder program is a simplified OEE-related indicator based on stop counts and the production count. A complete OEE calculation (Availability × Performance × Quality) needs planned production time, ideal cycle time, and good/reject part counts, and is identified as future scope.

CHAPTER 2: LITERATURE SURVEY & PROPOSED WORK

2.1 Introduction

OEE is used worldwide in discrete and process manufacturing to measure how well planned production time is used. It was popularised through the Total Productive Maintenance methodology and is now standardised in manufacturing KPI frameworks.

PLC-based systems are commonly used as data collectors because they sit directly on the machine and can read signals at every scan. The PLC data is then passed to an HMI, a SCADA system, or a plant database for display and analysis.

Traditional paper-based OEE tracking is slow and prone to errors, whereas PLC-based tracking records events automatically and consistently.

The proposed project uses Mitsubishi PLC programming software to develop and simulate an OEE monitoring program that counts production and stoppage events and calculates a result in PLC registers.

2.2 Literature Survey Table

The following table summarizes relevant technologies and concepts used in the project. These are background references, not claims that specific papers were reviewed.

Sr. No.	Technology / Reference	Application	Relevance to Project
1	Overall Equipment Effectiveness (OEE)	Measuring manufacturing productivity	Defines the performance indicator the project monitors.
2	Total Productive Maintenance (TPM)	Reducing equipment losses	Source of the loss categories: stops, speed loss, and defects.
3	PLC-based data collection	Industrial automation	Provides the counting and calculation platform.
4	Ladder logic programming	Implementation of control circuits	Used to build the counting and calculation rungs.
5	Special relays (SM402, SM412, SM400)	First-scan, 1 s clock, and always-ON signals	Used to initialise D20, time the counters, and run calculations.
6	HMI / SCADA	Displaying machine data	Used to show D0–D3 and D14 to operators.
7	Mitsubishi GX Works3	PLC programming and monitoring	Used to develop and test the ladder program.

2.3 Problem Definition

In many production lines, stoppages and idle periods are not recorded accurately, so the real productivity of a machine is not known. Without data, the causes of losses cannot be identified and improvements cannot be measured.

A monitoring system is therefore required that counts production activity and stoppage events automatically and calculates a performance value that can be displayed. The system must satisfy the following requirements:

1. Counting must be enabled only when the Start / Production ON input (X0) is active.
2. The Total Production Count (D0) should increase while production is enabled.
3. Idle events (D1), minor stops (D2), and major stops (D3) should be counted separately.
4. Each stop input should be counted once per press, using rising-edge detection.
5. The calculation should run continuously and should use the constant stored in D20.
6. The result must not be calculated when D0 is zero, to avoid division by zero.
7. A Reset input (X4) should clear all counters.

2.4 Feasibility Study

Technical feasibility: The system can be implemented using a PLC with digital inputs and standard ladder instructions: contacts, INC, MOV, MUL, DIV, and comparison contacts.

Operational feasibility: The operator uses five pushbuttons or field signals, making the system straightforward to operate.

Economic feasibility: PLC simulation allows the monitoring logic to be tested before connecting any hardware, reducing initial testing costs. An existing PLC on the machine can often be reused for monitoring.

Data-range feasibility: The calculation uses integer arithmetic, so the result is a whole number, and registers have a limited range. For real plant use, the data types and scaling must be reviewed.

2.5 Methodology Used

The project follows these steps:

1. **Requirement analysis:** Identify the production states and stoppage types to be monitored.
2. **I/O and register assignment:** Assign inputs X0–X4 and data registers D0–D3, D10, D12, D14, and D20.

3. **Ladder diagram design:** Develop rungs for initialisation, counting, reset, and calculation.
4. **Edge detection:** Use rising-edge contacts so that each stop press is counted once.
5. **Calculation design:** Implement D10, D12, and D14 with a divide-by-zero check.
6. **Simulation:** Run the ladder program using GX Works3 and GX Simulator.
7. **Testing:** Verify counting, reset, and calculation behaviour and prepare the HMI display plan.

CHAPTER 3: ANALYSIS AND PLANNING

3.1 Introduction

The analysis and planning stage identifies the monitoring requirements and defines the PLC logic needed to count events and calculate the result. The system is divided into input devices, PLC processing logic, and output/display values. The input devices provide production and stoppage signals, while the PLC processes them according to the ladder program. The outputs are the register values D0 to D3, D10, D12, D14, and D20, which can be shown on an HMI, SCADA screen, or PC. The program is tested in simulation before any connection to a physical machine is made.

3.2 Project Planning

Phase	Activity	Expected Deliverable
Phase 1	Requirement analysis	List of production states and losses
Phase 2	I/O and register assignment	PLC input and register table
Phase 3	DFD, block diagram, and flow chart	Design diagrams
Phase 4	Ladder diagram design and programming	Ladder program in GX Works3
Phase 5	Simulation	Verified register operation
Phase 6	Testing and documentation	Results and final report

3.3 Hardware and Software Requirements

Sr. No.	Component / Software	Purpose
1	Mitsubishi PLC (FX3U / iQ-F) or compatible simulator	Executes the ladder logic
2	Pushbuttons / field sensors (X0–X4)	Provide Start, Idle, Minor Stop, Major Stop, and Reset signals
3	24 V DC power supply	Powers the PLC and input devices
4	HMI / SCADA / PC	Displays counts and calculated values
5	MELSOFT GX Works3	Creates and edits the PLC program
6	GX Simulator3	Tests the program without physical hardware

Note: The supplied images show the ladder program in GX Works3 and the GX Simulator interface. The report does not assume that a physical machine has been connected or tested.

CHAPTER 4: DESIGN AND IMPLEMENTATION

4.1 Data Flow Diagram (DFD)

The Data Flow Diagram describes how data moves through the OEE Monitoring System. **Level 0** shows the whole system as one process (0 – OEE Monitoring System, PLC-based). The input devices X0 to X4 send input signals to the system, a 24 V DC power supply powers it, and the system outputs count data and calculated values (D0, D1, D2, D3, D10, D12, D14, and D20) to the HMI, SCADA, or PC.

Level 1 divides the system into six processes: 1.0 Count Total Production (D0), 2.0 Count Idle Events (D1), 3.0 Count Minor Stop Events (D2), 4.0 Count Major Stop Events (D3), 5.0 Reset Counts, and 6.0 Calculate OEE. The counts are held in a data store of PLC registers (D0, D1, D2, D3, and D20). Process 6.0 reads D3, D20, and D0 and calculates D10 = D3 × D20, D12 = D10 × 100, and D14 = D12 ÷ D0 if D0 > 0, otherwise D14 = 0.

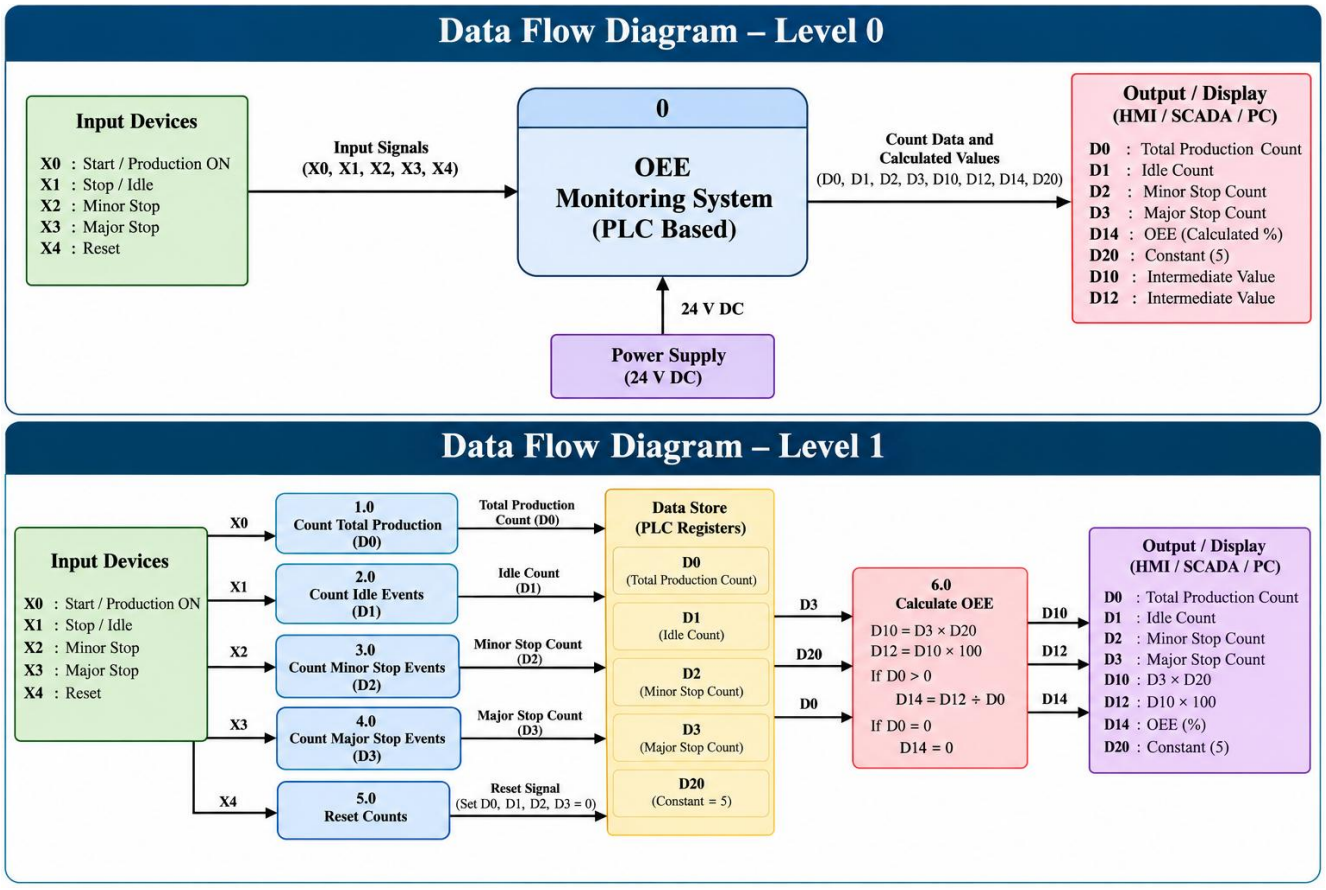


Fig. 4.1: Data Flow Diagram – Level 0 and Level 1

4.2 Block Diagram

The System Hardware and PLC Interfacing Block Diagram shows the physical interface of the system. The input pushbuttons are connected to PLC inputs and a 24 V DC supply powers the PLC and the input devices. The PLC executes the ladder program and the HMI/SCADA/PC displays the register values.

Input and register assignment

Address	Type	Description
X0	Input	Start / Production ON – enables counting
X1	Input	Stop / Idle – increments idle count
X2	Input	Minor Stop – increments minor stop count
X3	Input	Major Stop – increments major stop count
X4	Input	Reset – resets all counts (D0, D1, D2, D3)
D0	Register	Total Production Count
D1	Register	Idle Count
D2	Register	Minor Stop Count
D3	Register	Major Stop Count
D10	Register	Intermediate value, D10 = D3 × D20
D12	Register	Intermediate value, D12 = D10 × 100
D14	Register	Calculated result, D14 = D12 ÷ D0
D20	Register	Constant (K5), set on first scan

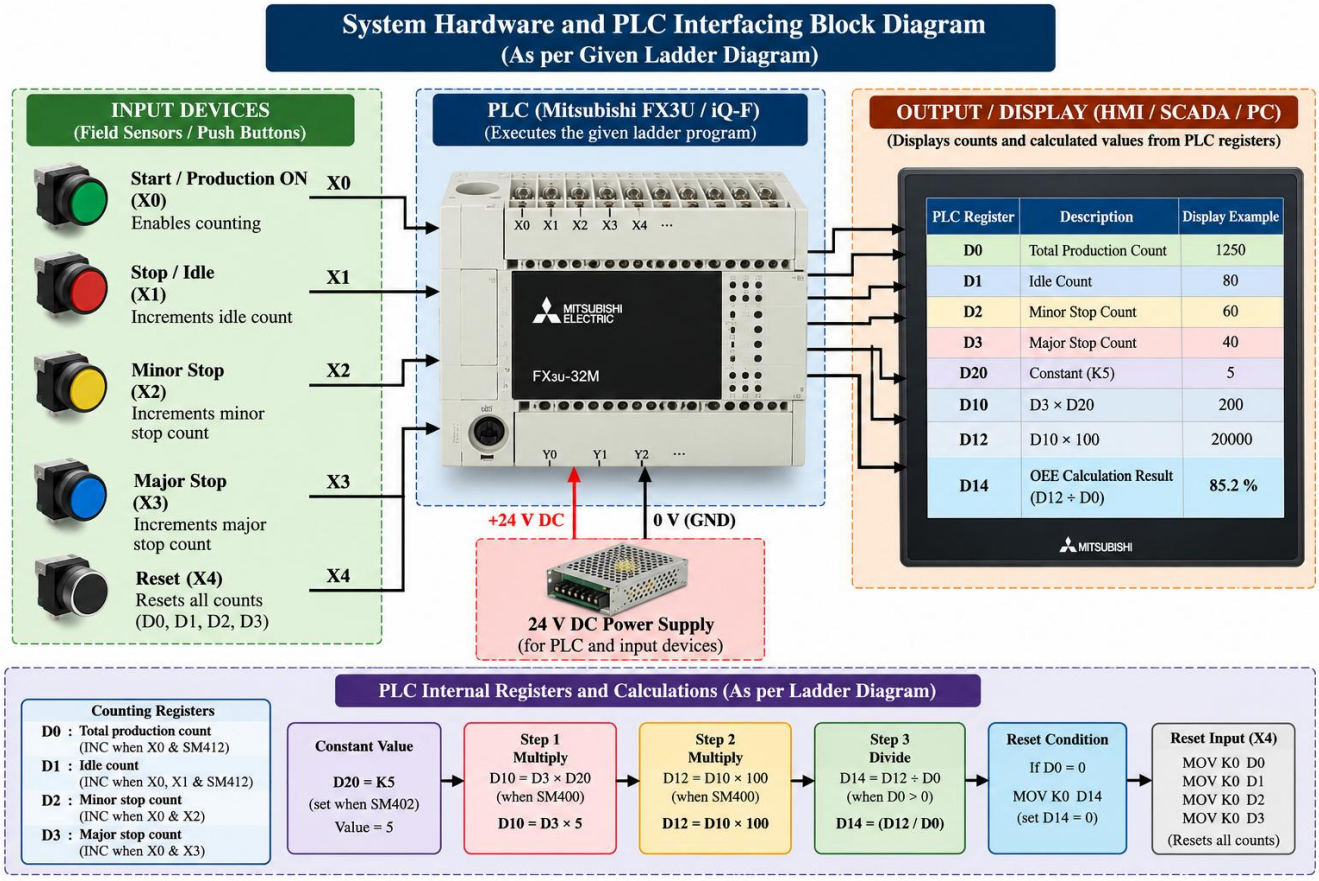


Fig. 4.2: System Hardware and PLC Interfacing Block Diagram

4.3 Flow Chart

The flow chart explains the working principle of the program. After the PLC powers on, the scan cycle starts and D20 is initialised to 5 once on the first scan (SM402). The program then continuously checks X0. If X0 is OFF, nothing is counted. If X0 is ON, the program checks X1, X2, and X3 in turn and increments D1, D2, or D3 when the corresponding input is active, and checks X4 to reset all counts.

The calculation section is executed every scan (SM400): D10 = D3 × D20 and D12 = D10 × 100 are calculated, and then D0 is checked. If D0 > 0, D14 = D12 ÷ D0 is calculated; otherwise D14 is set to 0. The scan cycle ends and repeats continuously.

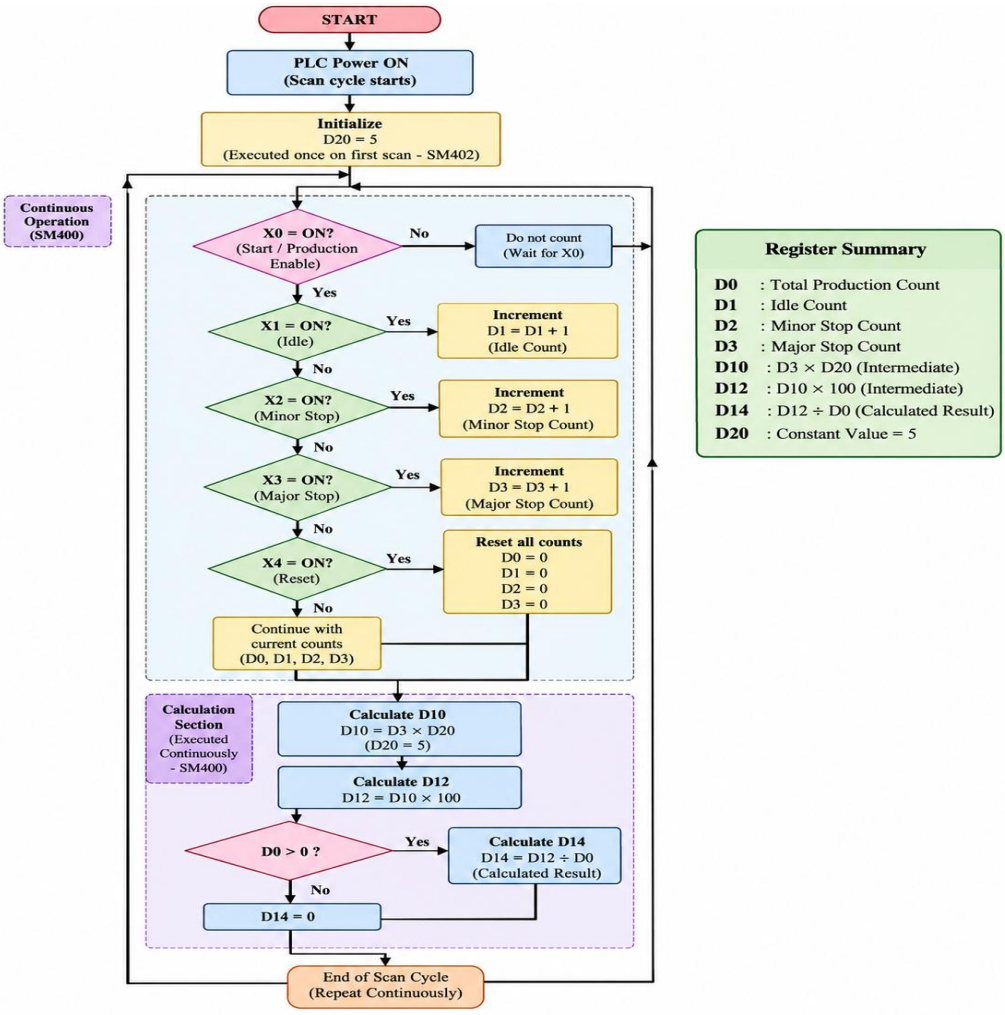


Fig. 4.3: Flow Chart of the OEE Monitoring Program

4.4 UML & PLC Ladder Diagram

The ladder diagram consists of 14 steps and ends with the END instruction. Its logic is explained below.

- **Rung 1, Initialisation:** SM402 (ON for one scan after RUN) executes MOV K5 D20, storing the constant 5 in D20.
- **Rung 2, Total production count:** X0 in series with the rising edge of SM412 (1 s clock) executes INC D0.
- **Rung 3, Idle count:** X0, X1, and the rising edge of SM412 execute INC D1.
- **Rung 4, Minor stop count:** X0 and the rising edge of X2 execute INC D2.
- **Rung 5, Major stop count:** X0 and the rising edge of X3 execute INC D3.

- **Rungs 6 to 9, Reset:** X4 executes MOV K0 into D0, D1, D2, and D3, one rung per register.
- **Rungs 10 and 11, Multiplication:** SM400 (always ON) executes MUL D3 D20 D10 and MUL D10 K100 D12.
- **Rung 12, Division:** The comparison contact D0 > K0 executes DIV D12 D0 D14.
- **Rung 13, Zero protection:** The comparison contact D0 = K0 executes MOV K0 D14.
- **Rung 14:** END.
- **Calculation:** D14 = (D3 × D20 × 100) ÷ D0 when D0 > 0, otherwise D14 = 0.

Because the SM412 contact is a 1-second clock, D0 and D1 advance once per second while their conditions are true, so they behave as time-based counts. D2 and D3 count events, one per press of X2 or X3.

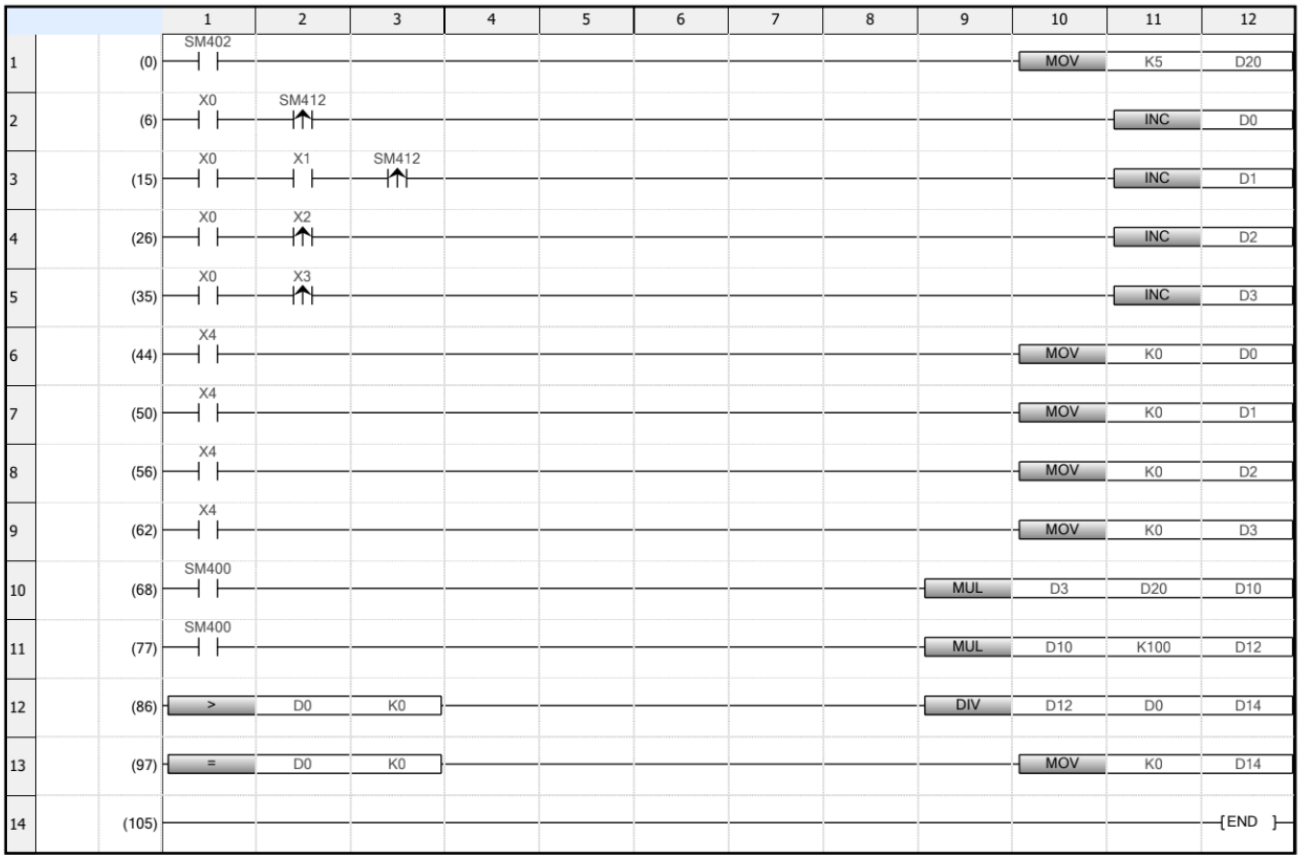


Fig. 4.4: PLC Ladder Diagram of the OEE Monitoring System

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Actual Results

The ladder program was loaded in MELSOFT GX Works3 and run with GX Simulator in RUN mode, as shown in Fig. 5.1. The monitor view shows that MOV K5 D20 was executed after RUN, so D20 holds 5, and that the counters D0, D1, D2, and D3 start from 0 before any input is operated. The comments "After RUN 1 Scan ON" and "1 s Clock" on the SM402 and SM412 contacts are visible in the program.

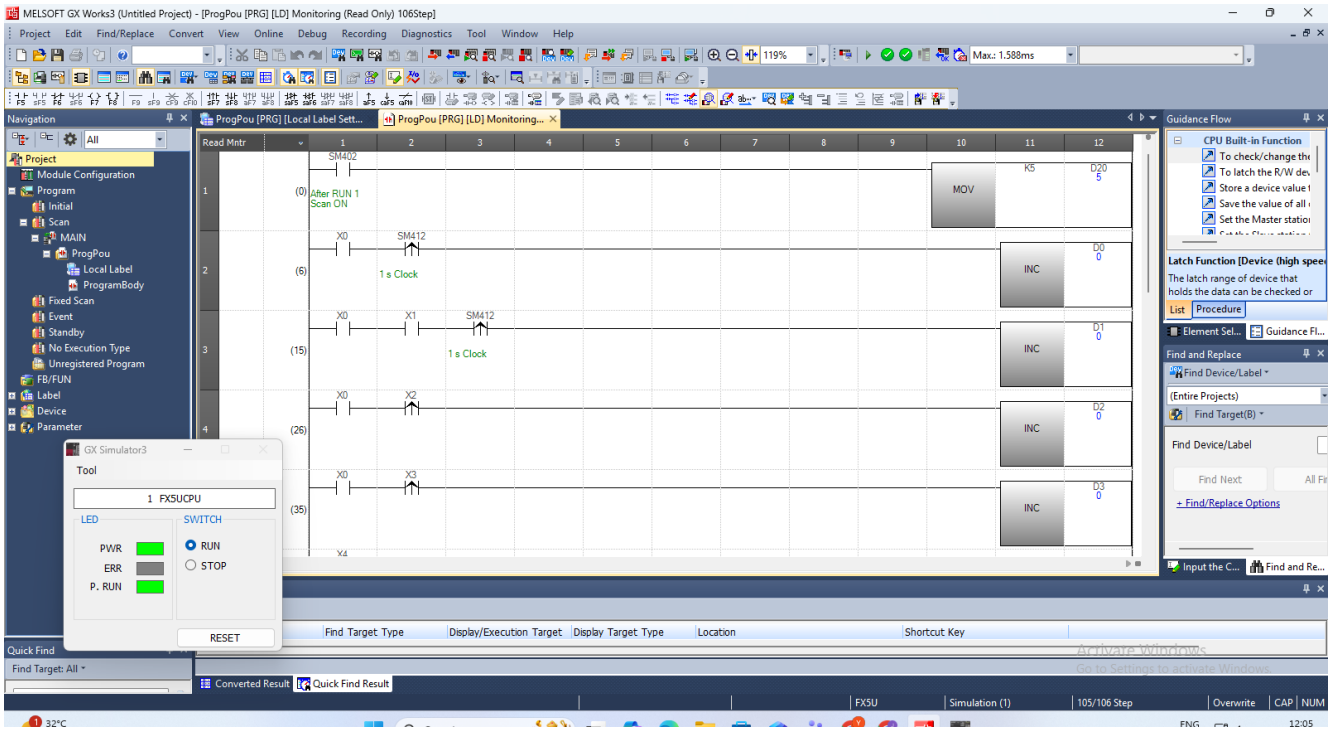


Fig. 5.1: Ladder Program Monitoring in GX Works3 with GX Simulator

The ladder program is intended to provide the following behavior during simulation.

Sr. No.	Test Condition	Expected Output
1	PLC set to RUN (first scan)	D20 = 5
2	X0 ON, X1, X2, X3 OFF, for 1 s clock cycles	D0 increases by 1 each second
3	X0 OFF	D0, D1, D2, and D3 do not change
4	X0 and X1 ON	D1 increases by 1 each second
5	X0 ON, X2 pressed once	D2 increases by 1
6	X0 ON, X3 pressed once	D3 increases by 1

Sr. No.	Test Condition	Expected Output
7	X2 or X3 held ON	Count increases only once (rising edge)
8	X4 pressed	D0, D1, D2, and D3 reset to 0
9	D0 = 0	D14 = 0 (no division)
10	D0 > 0	D14 = (D3 × 5 × 100) ÷ D0

These are expected behaviors from the intended control logic. Actual results should be recorded after performing the corresponding simulation tests.

Worked calculation

Using the example register values from the block diagram: D0 = 1250 and D3 = 40, with D20 = 5.

Register	Calculation	Result
D10	D3 × D20 = 40 × 5	200
D12	D10 × 100 = 200 × 100	20000
D14	D12 ÷ D0 = 20000 ÷ 1250	16

Note: The display example of 85.2 % shown in the HMI table of Fig. 4.2 is illustrative only. With the example counts above, the program as written gives D14 = 16. The DIV instruction gives an integer quotient, so decimal values are not shown.

Result

The designed ladder logic demonstrates counting of production activity and stoppage events, the use of special relays, arithmetic calculation using MUL and DIV, and divide-by-zero protection. The registers D0 to D3, D10, D12, D14, and D20 are available for display on an HMI, SCADA screen, or PC.

The calculated value D14 depends on the major stop count and the production count only. It is therefore a simplified indicator and not a full Availability × Performance × Quality OEE. This should be kept in mind when the result is interpreted.

5.2 Future Scope

1. **Complete OEE calculation:** Add planned production time, run-time timers, ideal cycle time, and good/reject counters to calculate Availability, Performance, and Quality.
2. **Floating-point result:** Use floating-point or scaled arithmetic to display decimal OEE percentages.
3. **Stop reason codes:** Record the reason for every stop for Pareto analysis.

4. **Real sensor integration:** Replace pushbuttons with photoelectric sensors, proximity sensors, and machine status signals.
5. **HMI integration:** Build a Mitsubishi HMI screen to display counts and OEE with trend graphs.
6. **SCADA and data logging:** Send data to a SCADA system or database for shift reports and history.
7. **Alarms:** Raise alarms when OEE falls below a target value or a major stop exceeds a set duration.
8. **IIoT connectivity:** Publish data to cloud dashboards for remote monitoring.

CHAPTER 6: CONCLUSION

The PLC-based OEE Monitoring System demonstrates the application of ladder logic in industrial production monitoring. The system uses five digital inputs to count production activity, idle events, minor stops, and major stops, and stores the values in data registers.

The ladder program uses special relays for first-scan initialisation, one-second timing, and continuous calculation. Rising-edge contacts ensure that each stop is counted once, and a Reset input clears all counters. Arithmetic instructions calculate the intermediate values D10 and D12 and the result D14, with a divide-by-zero check protecting the division.

The program was developed and tested using Mitsubishi MELSOFT GX Works3 and GX Simulator before connecting actual hardware. The Data Flow Diagram, block diagram, flow chart, and ladder diagram together document the design from system level down to program level. The project provides practical knowledge of PLC programming, register handling, arithmetic instructions, and monitoring in GX Works3.

Thus, the project establishes the basic logic required for an OEE monitoring system. For actual industrial implementation, a complete Availability × Performance × Quality calculation, real sensor signals, and an HMI/SCADA interface should be added.

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